

Ticket 00 Setup Check

Date: 2026-06-15

Scope

This is the Ticket 0 setup-only report for reports/03_solver/solver_implementation_tickets.md. No solver source code, Ticket 1 files, payoff utilities, Black-Scholes utilities, CN/PSOR code, boundary code, Greek code, dataset code, stress-map code, or neural-network code were created.

Source context checked:

- reports/03_solver/solver_implementation_tickets.md, Ticket 0.
- reports/00_planning/planning_report.md, workflow stages and decision gates.
- reports/03_solver/solver_validation_plan.md, sections 14 and 19.
- docs/student_handout_full_report.pdf, reproducibility guide.

1. Current Repo Structure

The repository is currently documentation-first. After this Ticket 0 report folder is created, the top-level structure is:

```
.
|-- README.md
|-- docs/
|   |-- README.md
|   |-- Student_Methodology_FreeBoundary_Risk_Surfaces.pdf
|   |-- student_handout_full_report.pdf
|-- reports/
|   |-- 00_planning/
|   |   |-- planning_report.md
|   |-- 01_literature/
|   |   |-- literature_map.md
|   |-- 02_math/
|   |   |-- formulation_note.md
|   |-- 03_solver/
|   |   |-- solver_implementation_tickets.md
|   |   |-- solver_validation_plan.md
|   |   |-- tickets/
|   |       |-- ticket_00_setup_check.md
```

No Python source files were found.

2. Required Directory Check

Directory	Exists now?	Notes
---	---	---
`src/`	No	Future source package location.
`tests/`	No	Future test location.
`experiments/`	No	Future validation script location.
`results/`	No	Future generated artifact location.

The solver validation plan lists future files under src/, experiments/, and results/, but it also says not to create those files during the planning stage. This report follows that rule.

3. Recommended Python Package and Import Convention

Recommended convention:

- Use a src layout with one explicit package namespace, preferably

src/american_risk_surfaces/.

- Put future modules under package subdirectories, for example

american_risk_surfaces.solvers, american_risk_surfaces.diagnostics, american_risk_surfaces.boundaries, and american_risk_surfaces.greeks.

- Use absolute imports in tests and experiments, for example:

```
from american_risk_surfaces.solvers.black_scholes import ...
```

Rationale:

- This is compatible with the student handout's PYTHONPATH=src convention.
- It avoids importing from bare top-level names such as solvers, which are more likely to collide

with unrelated packages.

- It keeps future Ticket 1 and later implementation import paths clear before code exists.

If the team chooses to keep the roadmap's shorter future paths such as src/solvers/..., that can work with PYTHONPATH=src, but it is less robust than a single project package namespace.

4. Dependency Status

No project dependency or packaging file was found. Specifically, no pyproject.toml, requirements*.txt, setup.py, setup.cfg, tox.ini, noxfile.py, pytest.ini, uv.lock, Pipfile, poetry.lock, or environment*.yml file was present.

Observed local runtime status:

```
| Runtime | Available packages checked |
| --- | --- |
| System `python3` 3.13.5 | `numpy`, `scipy`, `pandas`, and `matplotlib` are importable;
`pytest`, `reportlab`, `pypdf`, and `pdfplumber` are not importable. |
| Bundled Codex Python | `numpy`, `pandas`, `reportlab`, `pypdf`, and `pdfplumber` are
importable; `scipy`, `matplotlib`, and `pytest` are not importable. |
```

Observed PDF tooling:

- Bundled pdfinfo is available.
- Bundled pdftoppm is available.
- Bundled pdftotext is not available.

Conclusion: the repository currently has no declared reproducible dependency environment. Local machines may have enough scientific packages to start prototyping, but that is not a project-level setup.

5. Recommended Test Command

Because pytest is not installed in either checked Python runtime and the handout already gives a standard-library test command, the recommended immediate test command is:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src \  
python3 -m unittest discover -s tests
```

Recommended compile check once src/, tests/, and experiments/ exist:

```
PYTHONPYCACHEPREFIX=/private/tmp/codex_pycache PYTHONPATH=src \  
python3 -m compileall -q src tests experiments
```

Current result: there is no tests/ directory and no Python source. Existing test discovery is therefore not meaningful yet. python3 -m pytest --version failed because pytest is not installed. python3 -m compileall . completed successfully but found no project Python source to compile.

If the team wants pytest, approve a setup change first and declare it in project metadata. Then the future command can become:

```
PYTHONPATH=src python3 -m pytest tests -q
```

6. Setup Changes Needed Before Ticket 1

Yes, at least a small setup decision is needed before Ticket 1 implementation begins.

Recommended but not created in this Ticket 0 pass:

- Add a minimal pyproject.toml or requirements.txt.
- Decide whether the project uses unittest only or adds pytest.
- Create src/ and tests/ as part of the approved setup or Ticket 1 work.
- Prefer the package namespace american_risk_surfaces under src/.
- Declare numerical dependencies needed by future solver work, likely at least numpy; add scipy,

pandas, and matplotlib only if the implementation and validation scripts actually use them.

No setup files were created because the Ticket 0 instructions require proposing setup files in this report unless explicitly approved.

7. Next Recommended Action

Human review should approve one of these setup paths before Ticket 1:

1. Minimal standard-library testing path: create src/american_risk_surfaces/, tests/, and use unittest with the handout command.
2. Pytest path: create pyproject.toml, declare runtime and test dependencies, and use python3 -m pytest tests -q.

Recommended choice: start with the minimal standard-library testing path unless the team already wants pytest conventions. After that approval, Ticket 1 can add only payoff and European closed-form utility work with focused tests.

Inspection Commands Run

```

sed -n '1,220p' reports/03_solver/solver_implementation_tickets.md
pwd
ls -la
rg --files -g 'pyproject.toml' -g 'requirements*.txt' -g 'setup.py' -g 'setup.cfg' -g 'tox.ini' -g 'noxfile.py' -g 'pytest.ini' -g '.python-version' -g 'uv.lock' -g 'Pipfile' -g 'poetry.lock' -g 'environment*.yaml'
find . -maxdepth 2 -type d
rg --files
sed -n '1,220p' README.md
find . -maxdepth 1 -type d -name src -o -name tests -o -name experiments -o -name results
rg -n "(
###|###)|Gate|Stage|stage|gate|reproduc|environment|dependency|test|pytest|src|experiments|results" reports/00_planning/planning_report.md
rg -n "(
###|###)|14|19|reproduc|artifact|test|pytest|dependency|environment|src|experiments|results" reports/03_solver/solver_validation_plan.md
which python3
python3 --version
which pytest
find . -name '*.py' -type f
python3 -m pytest --version
python3 -m compileall .
/Users/xrx/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 -m
pytest --version
/Users/xrx/.cache/codex-runtimes/codex-primary-runtime/dependencies/bin/pdftools docs/student_handout_full_report.pdf
python3 -c "import importlib.util; ..."
/Users/xrx/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 -c "import importlib.util; ..."
sed -n '622,642p' reports/03_solver/solver_validation_plan.md
sed -n '767,780p' reports/03_solver/solver_validation_plan.md
sed -n '247,317p' reports/00_planning/planning_report.md
sed -n '399,444p' reports/00_planning/planning_report.md
/Users/xrx/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 -c "from pypdf import PdfReader; ..."
sed -n '1,220p' docs/README.md
git status --short
git branch --show-current
mkdir -p reports/03_solver/tickets

```